

Block Chain Technology			
Course Code:	22MCA343	Course Type	PCE
Teaching Hours/Week (L:T:P: S)	3:0:0:0	Credits	03
Total Teaching Hours	39	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
<u>Prerequisites :</u>			
Student must possess basic knowledge of internet, have Interest in learning Blockchain and Computer with Mac, Windows, or Linux OS			
Course Objectives:			
1.	To understand Block Chain, Blockchain Architecture and its Applications		
2.	To understand crypto currencies, Types and Applications		
3.	To understand Concept of Double Spending, Hashing, Mining, payment verification, Resolving Conflicts and Creation of Blocks		
4.	To understand Crypto currency wallets and conversion to Fiat Currency.		
5.	To Understand Smart contracts, usage, application, working principle, Law and Regulations		
UNIT - I			
Introduction to Blockchain, How Blockchain works, Blockchain vs Bitcoin, Practical applications, public and private key basics, pros and cons of Blockchain, Myths about Bitcoin.			07 Hours
UNIT - II			
Blockchain : Architecture, versions, variants, use cases, Life use cases of blockchain, Blockchain vs shared Database, Introduction to crypto currencies, Types, Applications.			08 Hours
UNIT - III			
Concept of Double Spending, Hashing, Mining, Proof of work. Introduction to Merkel tree, Privacy, payment verification, Resolving Conflicts, Creation of Blocks			08 Hours
Unit - IV			
Introduction to Bitcoin, key concepts of Bitcoin, Merits and De Merits Fork and Segwits, Sending and Receiving bitcoins, choosing bitcoin wallet, Converting Bitcoins to Fiat Currency.			08 Hours
Unit - V			
Introduction to Ethereum, Advantages and Disadvantages, Ethereum vs Bitcoin, Introduction to Smart contracts, usage, application, working principle, Law and Regulations. Case Study.			08 Hours
Course Outcomes: At the end of the course student will be able to			
1.	Understand and explain Block Chain, blockchain architecture and its applications		

2.	Understand and explain crypto currencies, types and applications													
3.	Understand and explain the concept of double spending, hashing, mining, payment verification, resolving conflicts and creation of blocks													
4.	Understand crypto currency wallets and conversion of crypto currency to fiat currency.													
5.	Understand and explain smart contracts, usage, application, working principle, law and regulations in relation to crypto currency.													
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes														
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO	PSO
↓ Course Outcomes													1	2
22MCA343.1	3	2		3	3				3			3	3	1
22MCA343.2	2	3		2	2				2			3	3	1
22MCA343.3	3	3		3	3				2			3	3	1
22MCA343.4	2	2		3	2				3			3	3	1
22MCA343.5	2	3		2	3				2			3	3	1
1: Low 2: Medium 3: High														
0														
TEXT BOOKS:														
1.	Arshdeep Bikramaditya Signal, Gautam Dhameja, “Beginning Blockchain: A Beginner's Guide to Building Blockchain Solutions”, APress													
2.	Bahga, Vijay Madiseti, “Blockchain Applications: A Hands-On Approach”. 31 January 2017													
3.	Melanie Swan, “Blockchain”, OReilly													
REFERENCE BOOKS:														
1.	Aravind Narayan. Joseph Bonneau, “Bitcoin and Crypto currency Technologies”, princeton													
2.	Arthu.T Books, “Bitcoin and Blockchain Basics: A non-technical introduction for beginners”													